# JUAN SEBASTIÁN SÁNCHEZ URBANO

## Industrial Automation Engineer · Machine Learning, Technical Sales, Power BI and AI

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LinkedIn · GitHub · Portfolio

## PROFESSIONAL PROFILE

Industrial Automation Engineer focused on the intersection of Machine Learning, Data Engineering and Consultative Sales. I connect plant operations and commercial strategy with Artificial Intelligence solutions and advanced analytics, to turn complex data into profitable sales and production decisions.

## TECHNICAL SKILLS

**Machine Learning & AI:** Machine Learning, Predictive modeling, Classification, Statistical analysis, EDA, AI agents, AI chatbots, Python, R.

**Sales & Business Intelligence:** Technical and commercial advisory, Sales analysis, Power BI, Advanced Excel, Dashboards, Data consolidation.

**Data engineering & Cloud:** ETL pipelines, Medallion architecture, Advanced SQL, MySQL, AWS, REST APIs, Data quality and cleaning.

**Automation & Industrial engineering:** n8n, Industrial automation, PLC programming, CNC and robotics, Real-time IoT sensors.

**Web development & deployment:** HTML, CSS, JavaScript, Bash, YAML, Git and GitHub, GitHub Actions, GitHub Pages.

**Methodology & Management:** PDCA (continuous improvement), Analytical problem solving, Results orientation, Effective communication.

## PROFESSIONAL EXPERIENCE

### Technical, Sales and Data Automation Advisor — Proeléctricos · Popayán, Cauca

*2019 – Nov 2025 · Mar 2026 – Sep 2026*

- I analyze and validate customers' technical and operational information to structure comprehensive solutions, ensuring the quality and traceability of commercial records.
- **Achievement:** I designed and implemented an AI chatbot for the web channel, integrated into an n8n automation pipeline that automatically extracts, validates and routes reports and information to the responsible areas.

### CNC and Robotics Area Lead (Production Data Analysis) — Metálicas Mundial · Candelaria, Valle del Cauca

*Oct 2025 – Jan 2026*

- I led the capture and analysis of critical production and quality-control variables, ensuring the reliability of manufacturing process data.
- **Achievement:** I identified bottlenecks and optimized cycle times by analyzing machinery operating data under agile continuous-improvement methodologies, increasing the area's productivity.

## FEATURED PROJECTS

### Immersive environments for phobia identification

*Bachelor's thesis · September 2025 · Unity, Meta Quest 2, Realidad virtual*

- I designed and programmed immersive environments in Unity for Meta Quest 2, focused on identifying phobias such as fear of fire and heights.
- **Impact:** I proposed a virtual-reality system that analyzes user reactions to controlled stimuli, for fear assessment and mental-health applications.

### Mobile robot control through virtual reality

*CINVESTAV, Mexico City · February 2026 · Unity, C#, ROS, Meta Quest 2*

- Development of a mobile-robot control system through virtual reality, using Unity and Meta Quest 2, in the Automatic Control Area of CINVESTAV.
- **Impact:** Integration of serial communication, IMU and LIDAR sensors, 3D modeling and robot control from the virtual-reality environment.

## EDUCATION

### Industrial Automation Engineering

*Universidad del Cauca · Graduated 2025*

### Complementary training

- Artificial Intelligence Bootcamp (MinTIC / Talento Tech) · Data Science, Python and Machine Learning (Platzi) Platzi profile